SADEEM FAHAD ALOTAIBI

sadeemfahd@outlook.sa 0547564759 Dhahran, Saudi Arabia Saudi

linkedin.com/in/sadeem-alotaibi-0b48092b5 github.com/sadeemfahd

SUMMARY

Software Engineering graduate from KFUPM with hands-on experience in full-stack development, cloud computing, and VR development. Skilled in Python, Java, JavaScript, Node.js, and Unity (C#), with practical experience in RESTful APIs, databases, cloud deployment, and interactive VR applications. Strong understanding of the Software Development Life Cycle (SDLC).

EDUCATION

Bachelor of Software Engineering, 2021 – 2026
King Fahd University of Petroleum and Minerals (KFUPM)

SKILLS

Programming Languages:
Python, Java

Data & Analytics:
Data Analysis, Machine Learning (KNN, SVM), Tableau, Hadoop, Microsoft Excel

Cloud & DevOps:
Google Cloud Platform (Cloud Run, API Gateway, Artifact Registry), Docker, Terraform, CI/CD, Git, GitHub

Web Development:
HTML, CSS, JavaScript, Node.js, Express.js, RESTful APIs

VR Development:
Unity (C#), XR Toolkit, Meta Quest, UI Interaction Systems

Databases & Tools:
MongoDB, Mongoose, Firestore

PROJECTS

VR-Integrated Flare System for Safe Management in Sour Gas Recovery Units (Senior Project)
Developed a VR-based SRU flare training system using Unity, featuring an interactive control room dashboard.
• Integrated engineering data from Aspen HYSYS → Excel → JSON → Unity for process monitoring.
• Implemented Blocked Outlet and Tube Rupture training scenarios with an interactive quiz and instant feedback.
• Tech: Unity (C#), Meta Quest, XR Toolkit, JSON, Tableau

Summrah Website — Full-Stack Web Development
Developed a full-stack web platform for preserving and documenting Saudi oral heritage, featuring user authentication, story management, feedback systems, and an interactive regional map.
• Implemented backend functionality and APIs using Node.js and Express.js with MongoDB and Mongoose for data storage and management.
• Awarded third place in the Web Development and Engineering category at the KFUPM Computing Project Expo.
Tech**:** Figma, HTML, CSS, JavaScript, Node.js, Express.js, MongoDB Atlas, Mongoose.

Smart Productivity Task Manager — Cloud Applications Engineering
• Built two cloud-based microservices for task management and analytics using Node.js, Express.js, and RESTful APIs.
• Implemented CRUD operations, API-key security, health checks, logging, and endpoint testing.
• Containerized the application with Docker and deployed it to Google Cloud Run.
• Provisioned GCP infrastructure with Terraform, including Firestore, API Gateway, and Artifact Registry.

Consortia Management System — Software Requirements Engineering
• Developed an SRS for a system managing KFUPM research consortia and industry partnerships.
• Defined stakeholder needs, user stories, functional/non-functional requirements, and system scope.
• Created UML models, UI prototypes, and requirements traceability and validation artifacts.

Sleep Health & Lifestyle Analytics — Data Analytics & Machine Learning
• Cleaned and analyzed healthcare data using Python and Pandas, including EDA and visualizations.
• Developed and compared ML classification models for sleep disorder prediction, achieving **91.15% accuracy** with a Decision Tree.
• Recognized as one of the **Best Projects** by the KFUPM Industrial & Systems Engineering Club.

EXPERIENCE

VR Development Intern, *VR House* 06/2025 – 08/2025 | Dhahran, Saudi Arabia
• Developed an interactive VR environment using Unity (C#) and Meta Quest.
• Created C# scripts for user interactions, lighting controls, and interactive features.
• Implemented VR navigation and movement using XR Origin and XR Toolkit.
• Integrated and optimized 3D assets, lighting, and rendering to enhance performance and immersion.
• Tested and deployed the VR application on Meta Quest.